

JavaScript Arrays & Objects with Functions - Practice Problems

1. Sum of Array Elements

Write a function that takes an array of numbers and returns the sum of all elements.

Example

```
sumArray([1, 2, 3, 4, 5]);  
// Output: 15
```

2. Find the Largest Number

Write a function that returns the largest number in an array.

Example

```
findLargest([12, 5, 30, 8, 20]);  
// Output: 30
```

3. Count Even Numbers

Write a function that counts how many even numbers are present in an array.

Example

```
countEven([1, 2, 4, 7, 8, 10]);  
// Output: 4
```

4. Get All Names

Given an array of user objects, return an array containing only the names.

Example

```
const users = [
  { id: 1, name: "Raj" },
  { id: 2, name: "John" },
  { id: 3, name: "Sam" }
];

getNames(users);
// Output: ["Raj", "John", "Sam"]
```

5. Find User by ID

Write a function that finds and returns a user object based on the given id.

Example

```
findUserId(users, 2);

// Output:
// { id: 2, name: "John" }
```

6. Filter Adults

Return only users whose age is 18 or above.

Example

```
const users = [
  { name: "Raj", age: 25 },
  { name: "John", age: 16 },
  { name: "Sam", age: 20 }
];

getAdults(users);

// Output:
// [
//   { name: "Raj", age: 25 },
```

```
// { name: "Sam", age: 20 }  
// ]
```

7. Double All Numbers

Return a new array where every number is doubled.

Example

```
doubleNumbers([2, 4, 6]);  
  
// Output:  
// [4, 8, 12]
```

8. Calculate Total Price

Given an array of products, calculate the total price.

Example

```
const products = [  
  { name: "Pen", price: 10 },  
  { name: "Book", price: 50 },  
  { name: "Pencil", price: 5 }  
];  
  
totalPrice(products);  
  
// Output: 65
```

9. Count Occurrences of a Value

Count how many times a specific value appears in an array.

Example

```
countOccurrences([1, 2, 2, 3, 2, 4], 2);  
  
// Output: 3
```

10. Add a New Property

Add a property called `isActive` with value `true` to every object.

Example

```
const users = [  
  { name: "Raj" },  
  { name: "John" }  
];  
  
addIsActive(users);  
  
// Output:  
// [  
//   { name: "Raj", isActive: true },  
//   { name: "John", isActive: true }  
// ]
```

11. Find the Most Expensive Product

Return the product object with the highest price.

Example

```
const products = [  
  { name: "Pen", price: 10 },  
  { name: "Book", price: 50 },  
  { name: "Laptop", price: 50000 }  
];  
  
mostExpensive(products);
```

```
// Output:  
// { name: "Laptop", price: 50000 }
```

12. Sort Users by Age

Return a new array sorted by age in ascending order.

Example

```
const users = [  
  { name: "Sam", age: 30 },  
  { name: "Raj", age: 20 },  
  { name: "John", age: 25 }  
];  
  
sortByAge(users);  
  
// Output:  
// [  
//   { name: "Raj", age: 20 },  
//   { name: "John", age: 25 },  
//   { name: "Sam", age: 30 }  
// ]
```

13. Convert Array to Object

Convert an array of users into an object using `id` as the key.

Example

```
const users = [  
  { id: 1, name: "Raj" },  
  { id: 2, name: "John" }  
];  
  
convertToObject(users);  
  
// Output:  
// {  
//   1: { id: 1, name: "Raj" },
```

```
// 2: { id: 2, name: "John" }  
// }
```

14. Count Products by Category

Return an object showing the number of products in each category.

Example

```
const products = [  
  { name: "Pen", category: "Stationery" },  
  { name: "Book", category: "Stationery" },  
  { name: "Phone", category: "Electronics" }  
];  
  
countByCategory(products);  
  
// Output:  
// {  
//   Stationery: 2,  
//   Electronics: 1  
// }
```

15. Student Average Marks

Given an array of student objects, calculate the average mark.

Example

```
const students = [  
  { name: "Raj", marks: 80 },  
  { name: "John", marks: 70 },  
  { name: "Sam", marks: 90 }  
];  
  
averageMarks(students);  
  
// Output: 80
```